

CSE113

Foundations of Computer Science

State University of New York, Korea

Why we need proof in CS?

- Reliability
- Security
- Optimization
- Limitation of Computation
- AI Safety

Final 2021

Problem 5. [5 points]

Prove using contradiction that the cube root of an irrational number is irrational.

Definitions and facts about numbers

Symbols

- Integers $\mathbb{Z}$
- Natural numbers $\mathbb{N}$
- Real numbers $\mathbb{R}$
- $|x|$
- sum Σ
- $a \mid b$
- $b \bmod a$

Formal definitions

- Even/Odd numbers
- Rational/Irrational numbers
- Prime/Composite numbers

Even/odd numbers

We say an integer n is even if: $\exists k \in \mathbf{Z}$ such that $n = 2k$

How can you define an odd number?

Rational/Irrational numbers

We say a real number r is rational if $\exists m, n \in \mathbf{Z}$ such that $r = n/m$
(and n and m have no common divisor).

Prime/Composite numbers

We say a natural number n is prime if $n > 1$, and

$$\forall r, s \in \mathbf{N}, n = rs \rightarrow (r = 1 \vee s = 1)$$

$$d \mid n$$

We say a non-zero integer d divides an integer n , if

$$\exists k \in \mathbf{Z}, \text{ such that } n = k * d.$$

Direct proof

Methods of mathematical proof

Statements	Method of proof
Proving existential statements (Disproving universal statements)	Constructive proof Non-constructive proof
Proving universal statements (Disproving existential statements)	Direct proof Proof by mathematical induction Well-ordering principle Proof by exhaustion Proof by cases Proof by contradiction

What can Direct Proof prove?

- For all x , $P(x)$
- For all x , $P(x) \rightarrow Q(x)$
- There exists x , $P(x)$

- **Prove: If p is an even number, then p^2 is an even number**
- Proof.
 - Assume p is an even integer. By definition of an even number, $p = 2k$ for some integer k
 - Squaring both sides of “ $p=2k$ ”, we get $p^2 = 4k^2$
 - Thus $p^2 = 2 (2k^2)$ which is twice an integer
 - Thus p^2 is even
- QED.

Even + odd = odd

Proposition

- Sum of an even integer and an odd integer is odd.

- Proof.
 - Suppose n is an even number, and m is an odd number, we need to show $n+m$ is odd
 - since n is an even number, $n = 2k$ for some integer k
 - since m is an odd number, $m = 2k'+1$ for some integer k'
 - Thus $n+m = 2(k+k')+1$ which shows $n+m$ is odd.
- QED.

n is odd $\Rightarrow n^2$ is odd

Proposition

- The square of an odd integer is odd.

- Proof.
 - Suppose n is an odd number. We want to show that n^2 is an odd number.
 - Since n is odd, $n = 2k+1$ for some integer k
 - $n^2 = 4k^2+4k+1 = 2(2k^2+2k) + 1$
 - Thus n^2 is odd
- QED.

If $a|b$ and $b|c$, then $a|c$

Proposition

- (Transitivity) For integers a, b, c , if $a|b$ and $b|c$, then $a|c$.

- Proof.
 - Suppose a, b, c are three integers and $a|b, b|c$.
 - Since $a|b$, we have $b = ak$ for some integer k
 - Since $b|c$, we have $c = bk'$ for some integer k'
 - Thus, $c = a(k \cdot k')$
 - Thus $a|c$.
- QED.

Summary

- Proof techniques — direct proof. **Commonly used for proving “for all x , $P(x) \rightarrow Q(x)$ ”.**
- Also used in many variants

Direct proof Variation 1

- How to prove “If A, then B”
 - Suppose A, ... Therefore B.

Direct proof Variation 2

- How to prove “for all real number x , $P(x)$ ”
 - Let x be a real number. ...Therefore $P(x)$.

Direct proof Variation 3

- How to prove “for all real number x , $P(x) \rightarrow Q(x)$ ”
 - Let x be a real number. Suppose $P(x)$ Therefore $Q(x)$.

Direct proof Variation 4

- How to prove “there exists x , $P(x)$ ”
 - Let x be <something you choose>. We have $P(x)$ holds.

Direct Proof Exercises

Problem 4. [5 points]

Prove that the sum of the squares of any two consecutive odd integers is even.

- Proof.
 - We need to prove the following:
 - for any integer n , $(2n+1)^2 + (2n+3)^2$ is even.
 - Let n be an arbitrary integer.
 - We have $(2n+1)^2 + (2n+3)^2 = 8n^2 + 20n + 2 = 2(4n^2 + 10n + 1)$ following algebraic Identities.
 - Therefore, $(2n+1)^2 + (2n+3)^2$ is even.
- QED.

Problem 5. Direct proof (points = 5)

Suppose a , b and c are integers. If $a^2|b$ and $b^3|c$, then $a^6|c$.

- Proof.
 - Let a , b , and c be three integers.
 - Suppose $a^2 \mid b$ and $b^3 \mid c$
 - By definition, we have $b = k a^2$ for some integer k , and $c = k' b^3$ for some integer k' .
 - Thus, $c = (k' k^3) a^6$
 - Therefore $a^6 \mid c$.
- QED

**Prove: For any natural number n , $n^2 + 3n + 2$
is composite**

For any integer x, y , if x is even, then xy is even.

Prove: there exist two irrational number r_1 , r_2 , such that $r_1 * r_2$ is a rational number.

**Prove: Suppose a is an integer. If $7|4a$,
then $7|a$.**